応用数学　第 3 章 §2　練習問題 2（p. 104）

【フーリエ変換と積分定理】

問題 1

次の関数のフーリエ変換を求めよ．

(1) $f(x)=\begin{cases}2 & (0\le x<2)\\ 1 & (2\le x<3)\\ 0 & (x<0,\ x\ge 3)\end{cases}$

(2) $g(x)=\begin{cases}1-x/2 & (0\le x<2)\\ 0 & (x<0,\ x\ge 2)\end{cases}$

解答:

(1) $F(u)=\displaystyle\int_0^2 2e^{-iux}\,dx+\int_2^3 e^{-iux}\,dx$

$=\dfrac{2(1-e^{-2iu})}{iu}+\dfrac{e^{-2iu}-e^{-3iu}}{iu}=\dfrac{2-e^{-2iu}-e^{-3iu}}{iu}=\dfrac{-i(2-e^{-2iu}-e^{-3iu})}{u}$

(2) $G(u)=\displaystyle\int_0^2 e^{-iux}\,dx-\frac{1}{2}\int_0^2 xe^{-iux}\,dx.$

第 1 項 $=\dfrac{1-e^{-2iu}}{iu}=\dfrac{-i(1-e^{-2iu})}{u}.$

第 2 項は部分積分で $\displaystyle\int_0^2 xe^{-iux}\,dx=-\frac{2e^{-2iu}}{iu}-\frac{1-e^{-2iu}}{u^2}.$

$G(u)=\dfrac{-i}{u}+\dfrac{1-e^{-2iu}}{2u^2}=\dfrac{-2iu}{2u^2}+\dfrac{1-e^{-2iu}}{2u^2}=\dfrac{1-2iu-e^{-2iu}}{2u^2}$

問題 2

次の関数について，(1) はフーリエ正弦変換，(2) はフーリエ余弦変換を求めよ．

(1) $f(x)=\begin{cases}\sin x & (|x|\le\pi)\\ 0 & (|x|>\pi)\end{cases}$

(2) $g(x)=\begin{cases}\cos x & (|x|\le\pi/2)\\ 0 & (|x|>\pi/2)\end{cases}$

解答:

(1) f は奇関数．$F_s(u)=2\displaystyle\int_0^\pi \sin x\sin ux\,dx$

$$=\int_0^\pi[\cos(1-u)x-\cos(1+u)x]\,dx$$

$=\left[\dfrac{\sin(1-u)x}{1-u}-\dfrac{\sin(1+u)x}{1+u}\right]_0^\pi=\dfrac{\sin(1-u)\pi}{1-u}-\dfrac{\sin(1+u)\pi}{1+u}$

$\sin(1\mp u)\pi=\pm\sin u\pi$ より

$$F_s(u)=\frac{2\sin u\pi}{1-u^2}$$

(2) g は偶関数．$G_c(u)=2\displaystyle\int_0^{\pi/2}\cos x\cos ux\,dx$

$$=\int_0^{\pi/2}[\cos(1-u)x+\cos(1+u)x]\,dx$$

$=\left[\dfrac{\sin(1-u)x}{1-u}+\dfrac{\sin(1+u)x}{1+u}\right]_0^{\pi/2}=\dfrac{\sin(1-u)\pi/2}{1-u}+\dfrac{\sin(1+u)\pi/2}{1+u}$

$\sin(1\mp u)\pi/2=\cos(u\pi/2)$ より

$$G_c(u)=\frac{2\cos(u\pi/2)}{1-u^2}\quad(u\neq 1)$$